

Algorithmic Transaction Cost Analysis

INTRODUCTION

Transaction cost analysis (TCA) has regained a new found interest in the financial community as a result of the proliferation of algorithmic trading. Portfolio managers and traders are using TCA to evaluate performance of brokers and their algorithms. Furthermore, TCA is used by portfolio managers to improve performance as part of their stock selection and portfolio construction process.

Currently, there are many investors who utilize TCA to select their trading algorithms and make informed trading decisions. Those investors who are not yet utilizing TCA as a decision-making tool are missing valuable opportunities to improve portfolio performance and increase returns.

TCA has evolved significantly over the last several years, though it is still commonly conceptualized as a vague and unstructured concept. The accompanying literature and research still remains muddled due to misrepresentation by many brokers, vendors, and industry participants. We set out to shed new light below.

In order to fully assist investors' algorithmic transaction cost performance, we have developed a framework that consists of pre-, intra-, and post-trade analysis. Our framework is based on an unbundling scheme where costs are classified by ten components and categorized by where they occur during implementation. This scheme is based on the work of Perold (1988) and Wagner and Edwards (1993), and has been described in *Journal of Trading*, "The Expanded Implementation Shortfall: Understanding Transaction Cost Components," Kissell (2006), and in *Optimal Trading Strategies* (2003). Madhavan (2000, 2002) provides a detailed investigation of financial literature discussing transaction cost components and is considered by many as the gold standard of TCA literature review.